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Extracting and reconstructing knowledge in materials science literature using large language models

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Knowledge reconstruction is required for both existing and new material discovery. However, this process faces significant challenges, especially in ensuring the completeness and logical consistency of extracted information. We present a generalized method that can be used to reconstruct knowledge from inorganic science literature regarding synthetic routes and properties. Using a one-prompt design, a comprehensive dataset was built with the GPT-4 model, which was subsequently used to fine-tune four large language models (LLMs), including LLaMA3-8B-instruct (Llama), Gemma-7B (Gemma), Phi3-mini-128k-instruct (Phi), GPT3.5-turbo-1106 (GPT). The fine-tuned models demonstrated a strong capability in reconstructing synthetic routes for selective catalytic reduction (SCR). The information extraction (IE) process achieved high performance on the task of material synthesis route extraction, with a precision of 0.928, a recall of 0.957, and an F1 score of 0.962. The fine-tuned models were then used for large-scale IE, extracting 48,925 entities from 2205 articles. Due to the JSON structure of the initial GPT-4-constructed dataset, the extracted data was stored in the same format, providing a valuable resource for further research. The four fine-tuned LLMs were transferred to five other domains, including lithium-ion batteries (Li-ion), hydrogen evolution reaction (HER), oxygen reduction reaction (ORR), oxygen evolution reaction (OER), and methanol steam reforming (MSR), to test the generalized performance of the fine-tuned LLMs. The results show that the fine-tuned LLMs have high transferability across domains and can be applied to a wide range of materials science tasks. Ultimately, a comprehensive knowledge graph was constructed by integrating multiple entities (nodes) including article metadata, catalysts, preparation methods, raw materials, synthesis steps, synthesis conditions and performance metrics. These nodes are interconnected through various relational edges (such as synthesis method, synthesis action, temperature, pressure, time, atmosphere, equipment, conversion, selectivity, activity) that represent explicit physicochemical dependencies, procedural parameters, and performance correlations. This structured framework enables the efficient exploration of intricate relationships within materials science literature, thereby facilitating the rapid identification of high-performance catalysts and the data-driven optimization of synthesis.

The synthesis of new materials is the key to promoting scientific and technological progress[1]. However, materials synthesis presents significant challenges. The complexity of materials science requires that researchers possess extensive foundational knowledge in physics and chemistry, as well as proficiency in advanced experimental and computational methods. Moreover, materials are influenced by various factors during synthesis process, such as temperature, pressure, and chemical environment. For instance, SCR catalysts are commonly prepared via impregnation, precipitation, and sol-gel methods, each differing in raw materials, synthesis procedures, and reaction conditions. Minor deviations, such as subtle

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variations in solvent ratios and dissolution sequences during raw material solutions preparation, may lead to structural differences. The order of steps, active component loading and support shaping, or the sequence of precursor addition during co-precipitation, directly impact catalyst efficiency. These factors can influence the performance of materials, complicating research efforts. As the complexity of material composition and microstructure increases, traditional material discovery methods, including experimental trial-and-error approaches and theoretical computational methods are increasingly exhibiting limitations in terms of development cost, efficiency, and time[2].

Data-driven artificial intelligence (AI) has developed rapidly and played an important role in various fields[3]. The advantages of AI technology in materials design are becoming increasingly prominent, and the integration of AI and big data is being called the "fourth industrial revolution"[4,5]. These capabilities are primarily driven by machine learning (ML), enabling data-driven pattern recognition and predictive modeling. Various ML techniques, including regression models[6–8], classification models[9,10], neural networks[11], reinforcement learning[12], and graph neural networks[13], have been widely applied in materials development such as lithium batteries[14,15], metal-organic frameworks[16,17], chemical reactions[18], and materials design[19]. The training of ML models relies on a vast amount of reliable and normalized scientific data, which are mainly derived from existing databases, experimental or computational results from researchers' laboratories, and scientific literature published by other researchers[20,21]. In particular, scientific literature usually contains rich information on detailed experimental procedures, conditions, and observations[22]. In addition, these sources contain the latest peer-reviewed research results and innovative methods, with high reliability and credibility. Therefore, effectively mining and organizing information from the materials literature is critical for advancing data-driven materials research. Unfortunately, manually extracting and reconstructing knowledge from scientific literature is a labor-intensive process, which hinders the efficient acquisition of valuable insights.

In recent years, natural language processing (NLP) techniques have been successfully applied to extract scientific literature data in the fields of chemistry[23,24], medicine, biology[25], and materials science[26], mining a large amount of excellent data. The application of NLP techniques has accelerated ontology development in materials science. Bayerlein et al.[27] employed an NLP-driven ontology development approach to provide a standardized semantic framework for the field of materials science microscopy, effectively addressing issues of terminology ambiguity and retrieval efficiency, and advancing the application of FAIR (Findability, Accessibility, Interoperability, Reusability) data management principles in experimental science. Traditionally, specialized NLP methods including dictionary matching, rules, ML and deep learning models have been employed to solve this issue related to IE and semantic standardization. Liu et al.[28] utilized rule-based and domain dictionaries framework to extract valuable domain knowledge from abstracts of scientific literature, with a particular focus on Ni-based single crystal superalloys. Li et al.[23] designed rules to extract catalyst synthesis data from scientific literature, specifically focusing on Pd-based methanol steam reforming. The extracted data was used to train ML models, which were designed to recommend optimal synthesis routes. Liu et al. and Cruse et al. carried out extensive research, training BiLSTM-CRF models based on transformer architectures, to extract synthesis procedures for solution-based inorganic materials[29], and gold nanoparticles[30]. Additionally, ML models such as MatSciBERT[31] and ChemBERT[32] have been trained on annotated data to extract specific chemical information. These approaches can be labor-intensive and necessitate expertise in coding, computer science, and data science. Furthermore, they are less generalizable, requiring redesign or training when the target changes.

LLMs have demonstrated excellent capabilities in NLP to extract information from literature, understand and generate chemical data. Jablonka et al.[24] fine-tuned the GPT-3 model to perform tasks such as molecular property prediction, reaction yield estimation, and even inverse molecular design through natural language question. Zheng et al.[33] used prompt engineering techniques to guide ChatGPT in mining the scientific literature for the synthesis of metal-organic frameworks with information on metal sources, linkers, solvents, reaction temperatures, and times. Polak et al.[34] used conversational ChatGPT to extract materials properties such as the critical cooling rates of metallic glasses and yield strength of high entropy alloys in the form of the triplet material, value, unit. Walker et al.[35] proposed a method using GPT-3 model to extract structured multi-step seed-mediated growth procedures for gold nanorods from scientific text by utilizing a formulated synthetic template containing seed solution, growth solution and generated nanorods. Pei et al.[36] demonstrated that LLMs can expedite the materials design process and enable the inclusion of holistic criteria in the domain of alloy design and manufacturing. Dagdelen et al.[37] also used LLMs (GPT-3, Llama-2) to perform IE tasks: dopant-host systems, metal-organic frameworks, and general materials information. Their approach focused on the automatic extraction of entities such as materials, dopant-host relationships, structure components, phases, morphologies, and applications. Venugopal et al.[38,39] constructed MatKG and MatKG-2 knowledge graphs by extracting information such as materials, properties, applications, and characterization methods from abstract of literature, using BERT-based approaches for MatKG and LLMs (Llama2 7B and Bloom 1B/7B) for MatKG-2. Previous studies have successfully leveraged LLMs or neural network models to extract entities and relationships. Few studies[37] have addressed the comprehensive extraction of material names, synthesis details, material performance, source metadata, and information with strong logical relationships from the full text of scientific literature. Existing materials science information extraction studies focus on structured data acquisition[37,40–42], but differ from this work in LLM strategy, extracted information, cross-domain capability, and output format. These distinctions highlight the novelty of our universal framework for inorganic catalytic materials.

In this work, we present a method that leverages LLMs for comprehensive knowledge reconstruction from materials science literature. This approach enables detailed analysis of each step and condition within

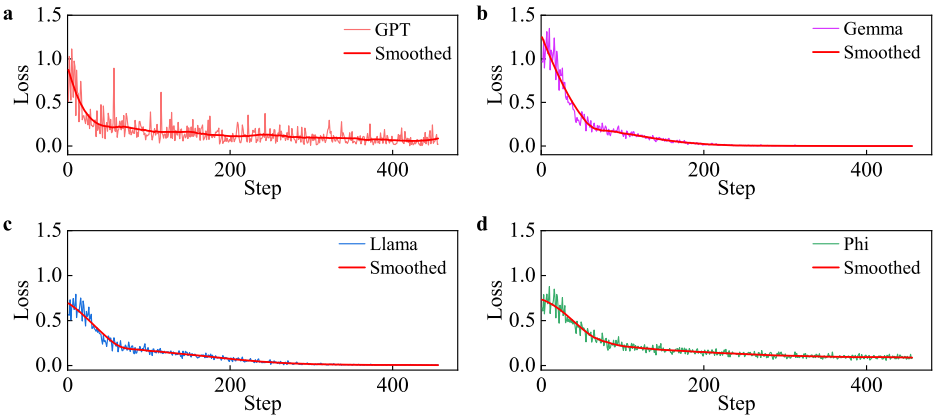


Fig. 1 | Loss curves during the fine-tuning process of LLMs. a Loss of GPT model. **b** Loss of Gemma model. **c** Loss of Llama model. **d** Loss of Phi model. All smoothed curves are represented in bright red.

Input:

LaFeO3samples were prepared by a citric acid sol–gel method. Equal amounts of lanthanum nitrate and ferric nitrate were dissolved in deionized water and mixed together, and then 120% (molar ratio) of citric was added into the solution and stirred for 30min. The solution was evaporated at 80°C until it became viscous. The obtained sample was dehydrated at 120°C for 12h, and then calcined in airflow at 750°C for 4h. LaxCe1−xFeO3samples were synthesized by the same way. The proportions of lanthanum nitrate and ceria nitrate varied according to the value of x in each LaxCe1−xFeO3sample. This process is similar to the LaFeO3synthesis process, except the addition of ceria nitrate in the first step.

Output:

GPT

{'Catalyst1': {'product': 'LaFeO3samples', 'method': 'citric acid sol\u2013gel method', 'raw material': ['Lanthanum nitrate', 'Ferric nitrate', 'Citric', 'Deionized water'], 'step': {'dissolve': {'reagents': ['Lanthanum nitrate', 'Ferric nitrate', 'Deionized water']}, 'mix': [], 'add': {'reagents': '120% citric', 'time': '30min'}, 'stir': {'time': '30min'}, 'evaporate': {'temperature': '80\u00b0C'}, 'dehydrate': {'temperature': '120\u00b0C', 'time': '12h'}, 'calcined': {'temperature': '750\u00b0C', 'time': '4h', 'atmosphere': 'airflow'}}}, 'Catalyst2': {'product': 'LaxCe1\u2212xFeO3samples', 'method': 'citric acid sol\u2013gel method', 'raw material': ['Lanthanum nitrate', 'Ceria nitrate', 'Ferric nitrate', 'Citric', 'Deionized water'], 'step': {'dissolve': {'reagents': ['Lanthanum nitrate', 'Ceria nitrate','Ferric nitrate', 'Deionized water']}, 'add': {'reagents': '120% citric'}, 'stir': {'time': '30min'}, 'evaporate': {'temperature': '80\u00b0C'}, 'dehydrate': {'temperature': '120\u00b0C', 'time': '12h'}, 'calcined': {'temperature': '750\u00b0C', 'time': '4h', 'atmosphere': 'airflow'}}}}

Gemma

{'Catalyst': {'product': {'name': 'LaFeO3', 'denoted': 'LaxCe1\u2212xFeO3'}, 'method': 'citric acid sol\u2013gel method', 'raw material': ['lanthanum nitrate', 'ferric nitrate', 'ceria nitrate'], 'step': {'dissolve': {'reagents': ['lanthanum nitrate', 'ferric nitrate', 'deionized water']}, 'mix': {'reagents': ['lanthanum nitrate', 'ferric nitrate', 'deionized water']}, 'add': {'reagents': ['120% citric acid'], 'ratio': '120% (molar ratio)'}, 'stir': {'time': '30min'}, 'evaporate': {'temperature': '80\u00b0C', 'condition': 'until the solution became viscous'}, 'dehydrate': {'temperature': '120\u00b0C', 'time': '12h'}, 'calcined': {'temperature': '750\u00b0C', 'time': '4h', 'atmosphere': 'airflow'}}}}

Llama3

{'Catalyst': {'product': {'name': 'LaFeO3', 'denoted': 'LaFeO3'},'method': 'citric acid sol\u2013gel', 'raw material': ['lanthanum nitrate', 'ferric nitrate', 'citric acid'],'step': {'dissolve': {'reagents': ['lanthanum nitrate', 'ferric nitrate', 'deionized water']},'mix': {}, 'add': {'reagents': 'citric acid'},'stir': {'time': '30min'}, 'evaporate': {'temperature': '80\u00b0C'}, 'dehydrate': {'temperature': '120\u00b0C', 'time': '12h'}, 'calcined': {'temperature': '750\u00b0C', 'time': '4h', 'atmosphere': 'airflow'}}}, 'LaxCe1\u2212xFeO3': {'product': {'name': 'LaxCe1\u2212xFeO3', 'denoted': 'LaxCe1\u2212xFeO3'},'method': 'citric acid sol\u2013gel', 'raw material': ['lanthanum nitrate', 'ceria nitrate', 'citric acid'],'step': {'dissolve': {'reagents': ['lanthanum nitrate', 'ceria nitrate', 'deionized water']},'mix': {}, 'add1': {'reagents': 'lanthanum nitrate'}, 'add2': {'reagents': 'ceria nitrate'},'stir': {'time': '30min'}, 'evaporate': {'temperature': '80\u00b0C'}, 'dehydrate': {'temperature': '120\u00b0C', 'time': '12h'}, 'calcined': {'temperature': '750\u00b0C', 'time': '4h', 'atmosphere': 'airflow'}}}}

phi3

{'Catalyst': {'product': {'name': 'LaFeO3', 'denoted': 'LaFeO3'},'method': 'citric acid sol\u2013gel method', 'raw material': ['lanthanum nitrate', 'ferric nitrate', 'citric'],'step': {'dissolve': {'reagents': ['lanthanum nitrate', 'ferric nitrate', 'deionized water']},'mix': {}, 'add': {'reagents': 'citric', 'time': '30min'}, 'evaporate': {'temperature': '80\u00b0C'}, 'dehydrate': {'temperature': '120\u00b0C', 'time': '12h'}, 'calcined': {'temperature': '750\u00b0C', 'time': '4h'}}}}

Fig. 2 | Example of inputs and outputs of the fine-tuned LLMs. The results of extracting information about the catalyst synthesis routes using four LLMs: GPT, Gemma, Llama3, and Phi3.

synthesis routes, precisely identifying hidden critical details and deep logical relationships. One-shot prompt was designed and a dataset was constructed with GPT-4 and human feedback, which was used to fine-tune four LLMs (LLaMA3-8B-instruct, Gemma-7B, Phi3-mini-128k-instruct, GPT3.5-turbo-1106). These models were then adopted to perform large-scale IE tasks. The extracted information was stored in structured JSON format, providing a valuable resource for future research. Applying this method, the synthesis routes and extracted corresponding performance data for SCR was successfully reconstructed. When evaluated across five domains (Li-ion, HER, ORR, OER, and MSR), our fine-tuned models consistently achieved high accuracy in capturing complex synthesis conditions and relational logic. This approach not only provides a scalable solution for literature-based knowledge reconstruction but also delivers a reusable structured resource for accelerated materials discovery.

Result and discussion

Dataset construction

The specifics of IE tasks are presented in the Methods section. Briefly, the synthesis tasks involve identifying products, synthesis methods, raw materials, and synthesis steps, and additional relevant information from text. Each base LLM was fine-tuned for task to follow a specific schema that defines the target entities, relevant relationships, and required format.

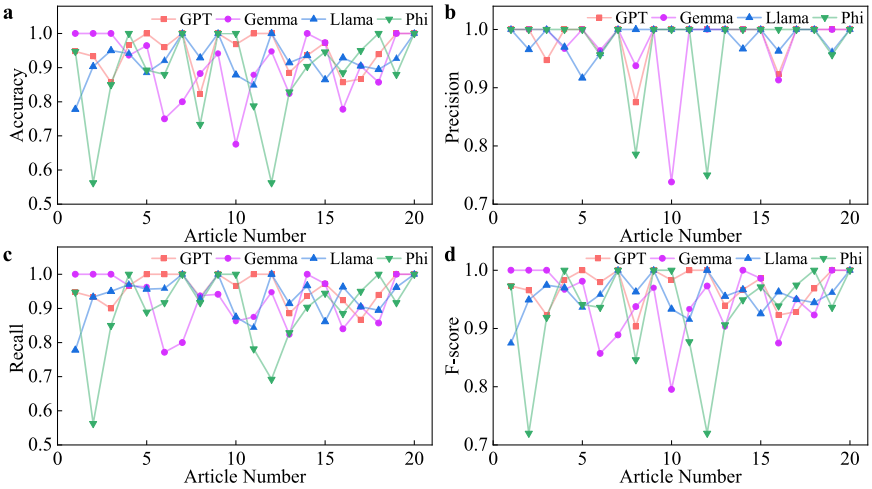


Fig. 3 | Test performance of GPT, Gemma, Llama and Phi in extracting catalyst synthesis routes on 20 Articles. The average accuracy **a**, precision **b**, recall **c**, F1 score **d** of four LLMs.

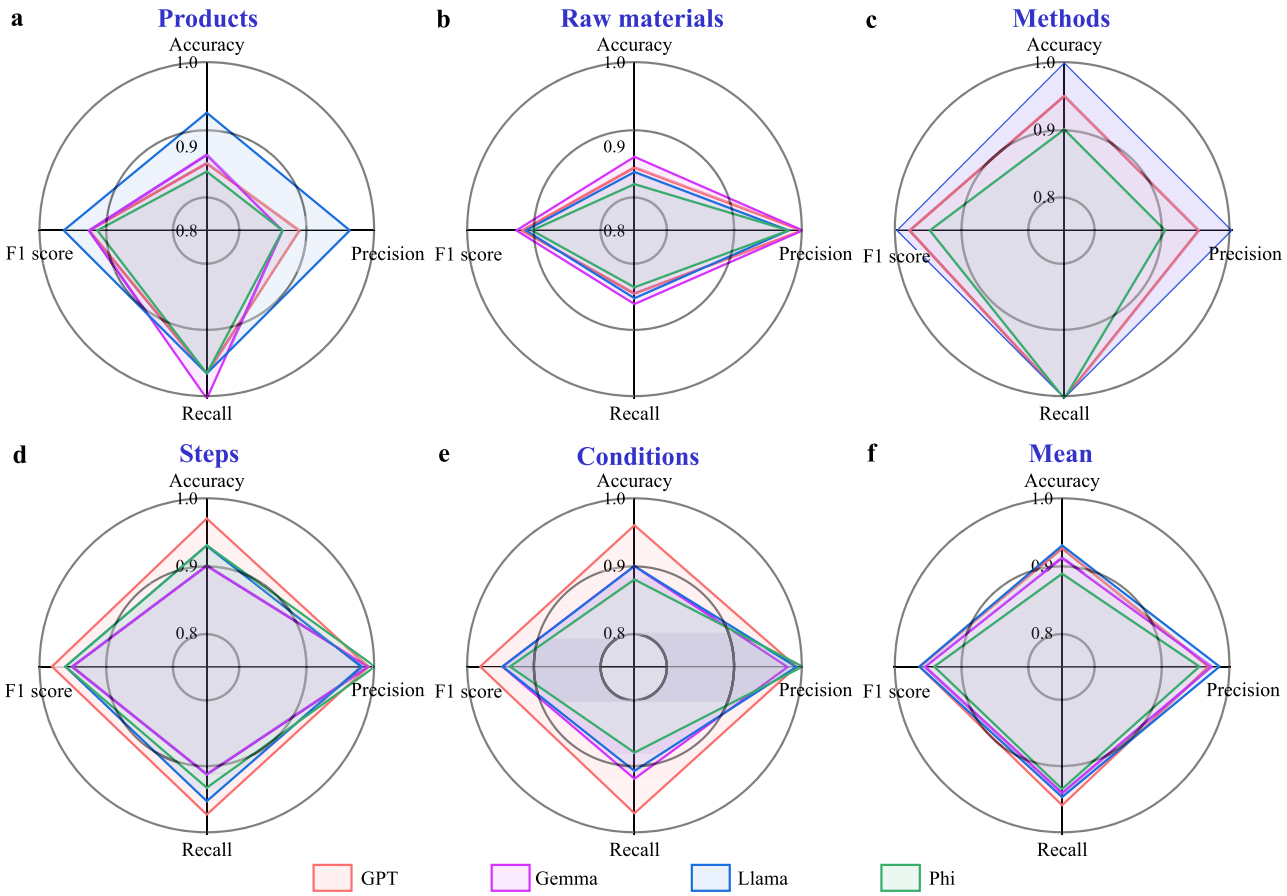


Fig. 4 | Performance of GPT, Gemma, Llama and Phi on five tasks. a The task of extracting product. **b** The task of extracting the raw materials in the synthesis process. **c** The task of extracting synthesis method. **d** The task of extracting synthesis steps. **e** The task of extracting the synthesis condition. **f** The average of the five tasks.

The one-shot prompt (described in Method) was designed to instruct GPT-4 to extract the synthesis route from the paragraph describing the material synthesis. Using the prompt, synthesis paragraphs were input to GPT-4 model, which extracted corresponding synthesis information, detailing each step, conditions, and materials involved. Three domain experts manually reviewed the GPT-4 generated content according to the output template, corrected the identified errors, and resolved discrepancies through discussion and negotiation to ensure annotation consistency and reliability. This iterative workflow, involving synthesis paragraph input, synthesis route generation, and manual review, improved prompt design with each iteration.

With this methodology, a work-cleaned dataset capturing detailed and accurate material synthesis routes was created, which was used to fine-tune LLMs. The work-cleaned dataset consists of two key components: synthesis paragraphs and the corresponding synthesis routes information. Each synthesis paragraph describes the detailed procedures and conditions for material synthesis. The dataset comprises a total of 156 synthesis paragraphs, containing 1330 sentences and 35855 tokens. Additionally, the dataset includes 156 synthesis routes, detailing raw materials, products, methods, and relevant information for each step, and encompasses 6337 entities. The dataset provides a comprehensive and structured representation of the synthesis process.

The OMS-2 samples were synthesized by the hydrothermal method (Wang et al., 2015). In a typical synthesis, MnAc2·4H2O (3.668 g) was dissolved in deionized water (35 ml), then glacial acetic acid (2.5 ml) was added to the above solution to form an acidic environment under stirring. KMnO4(2.168 g) was dissolved in deionized water (35 ml) and slowly added to the above mixed solution, then stirred for 40 min. The mixture was maintained at 100 °C in a Teflon-lined stainless-steel autoclave for 24 h, and the black brown slurry was cooled to room temperature. Finally, the slurry was rinsed with deionized water, then dried at 110 °C for 12 h. The OMS-2 material obtained by the above preparation method was called OMS-2-Ac. The OMS-2-SO4material was prepared in the same way, but with manganese sulfate [MnSO4] as Mn2+precursors.

{"Catalyst": {"product": {"name": "OMS-2-Ac", "denoted": "OMS-2-Ac"}, "method": "hydrothermal method", "raw material": ["MnAc2\u00b74H2O", "KMnO4", "glacial acetic acid", "MnSO4"],"step": {"dissolve": {"reagents": ["MnAc2\u00b74H2O", "deionized water (35ml)"], "instrument": "beaker"}, "add": {"reagents": ["glacial acetic acid (2.5ml)"], "instrument": "beaker"}, "dissolve2": {"reagents": ["KMnO4", "deionized water (35ml)"], "instrument":"beaker"},"stir": {"time": "40min"},"maintain": {"temperature": "100\u00b0C", "instrument": "Teflon-lined stainless-steel autoclave", "time": "24h"}, "cool": {"temperature": "room temperature"}, "rinse": {"reagents": "deionized water"}, "dry": {"temperature": "110\u00b0C", "time": "12h"}}}}

The catalyst was prepared by iron nitrite (Fe(NO3)3·9H2O, AR) produced by Sinopharm Chemical Reagent Co., Ltd. 10g Fe(NO3)3·9H2O was mixed in the 100mL deionized water under agitating with magnetic stirrer at the temperature of 80°C until the solution was dried. The obtained compounds were firstly dried at 120°C for 4h followed by calcinations at 550°C in air for 5h.

{"Catalyst": {"product": {"name": "catalyst", "amount": " ", "purity": "AR"}, "method": "prepared", "raw material": "Fe(NO3)3H2O", "step": {"mix": {"reagents": ["Fe(NO3)3\u00b79H2O", "100mL deionized water"]}, "dry": {"temperature": "120\u00b0C", "time": "4h"}, "calcinations": {"temperature": "550\u00b0C", "time": "5h", "atmosphere": "air"}}}}

factuality hallucination faithfulness hallucination

Fig. 5 | The different definitions of hallucinations. Blue text in the LLM's output indicates factuality hallucination, while red text indicates faithfulness hallucination.

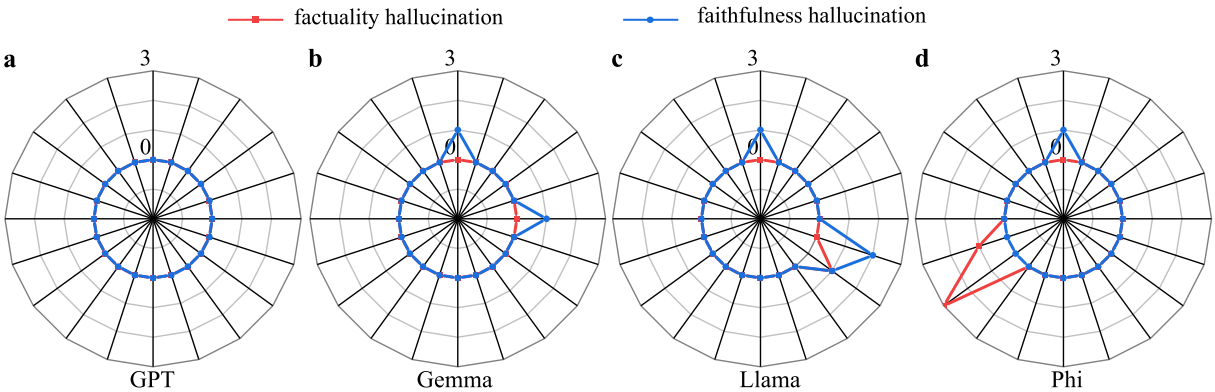


Fig. 6 | Evaluation of hallucination for four LLMs. The frequency of hallucinations in GPT model **a**, Gemma model **b**, Llama model **c**, Phi model **d**. The red lines indicate factuality hallucination and blue lines indicate faithfulness hallucination.

Fine-tuning performance

The described synthetic template training dataset was used to fine-tune the LLMs to reproduce synthetic templates in unstructured text. The parameters for the four LLMs' fine-tuning process were described in Methods. In this study, the Llama, Gemma, and Phi models were fine-tuned using supervised fine-tuning (SFT) with Low-Rank Adaptation (LoRA)[43] technology, which updates only the adapter weights while freezing the original pre-trained model parameters. The fine-tuning process for each model was completed in ~20 min on a single NVIDIA 4090 GPU, with peak memory usage around 23 GB. For the GPT model, online fine-tuning was performed, which required ~10 min without consuming local computational resources. The detailed procedure for the fine-tuning process is provided in the Supplementary Methods. Fine-tuning improves on few-shot learning by training on many more examples than can fit in the prompt, achieving better results on a wide number of tasks. Once fine-tuned, the model can generate responses without examples in the prompt.

The steps and loss graph for each model, as shown in Fig.1, illustrates the training loss against the training steps. This graph depicts the reduction in loss as training progresses, with each model demonstrating a steady decline in loss, indicating effective learning and model improvement over the epochs.

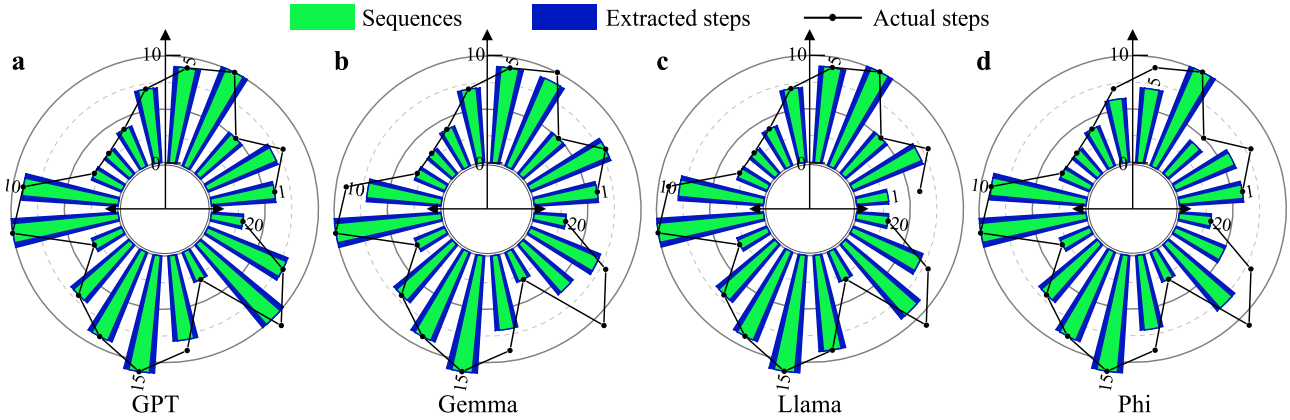


Fig. 7 | Evaluation of four LLMs on the completeness and logic of the extraction synthesis step. a GPT model. **b** Gemma model. **c** Llama model. **d** Phi model.

Input:

A series of Ce-Ti mixed-oxide catalysts were prepared by the sol–gel method for selective catalytic reduction (SCR) of NO with ammonia as reductant. These catalysts were characterized by XRD, BET, and XPS techniques. The experimental results show that the best Ce-Ti mixed-oxide catalyst yielded 98.6% NO conversion, and 100% N2 selectivity at typical SCR reaction temperatures (300–400°C) and the high gas hourly space velocity of 50,000h−1……

Output:


```
{
 "content": {
  "catalysts": "Ce-Ti mixed-oxide catalysts",
  "activity": "high",
  "conversion": "98.6%",
  "selectivity": "100%"
 },
 "role": "assistant"
}
```


Fig. 8 | Material property extraction. An example of input and output for extracting material properties from LLM.

Evaluation

Named entity recognition (NER) task was performed using the fine-tuned LLMs on the test datasets obtained from 20 articles and the performance of the fine-tuned models was evaluated. The Digital Object Identifier (DOIs) of 20 test articles are available in Supplementary Data (Supplementary Table 1). An example of inputs and outputs of different fine-tuned LLMs is shown in Fig. 2. The performance of four models (GPT, Gemma, Llama, and Phi) was evaluated in extracting different tasks: Products, Raw Materials, Methods, Steps, and Conditions. The evaluation metrics included accuracy, precision, recall, and F1 score.

To assess the overall performance of each model, we counted the average of the evaluation metrics of the 20 articles in the five tasks and displayed them in Fig. 3. Detailed results of the four models are shown in Supplementary Data (Supplementary Table 2-5). The results show that Phi and Gemma models fluctuates a lot, and the GPT-3.5 and Llama models are more stable in the test results. The GPT model performs the best in the extraction "step" and "condition" tasks out of the five tasks.

The performance of the four models on five types of tasks were evaluated, with results presented in Fig. 4 and detailed values in Supplementary Data (Supplementary Table 6). For the task of extracting "Products", the Llama model demonstrated superior performance, achieving the highest accuracy (0.944), precision (0.971), and F1 score (0.971). In the extraction of "Raw Materials", GPT performed best with an accuracy of 0.873, precision of 1.000, and F1 score of 0.932. For the task of extracting "Methods", both the Llama and Gemma models achieved perfect scores of 1.0 for accuracy, precision, recall, and F1 score, demonstrating exceptional performance. The GPT and Phi models also performed well, with recall and F1 score exceeding 0.9. It is noteworthy that the Gemma and Llama models achieved very similar performance metrics, resulting in their data markers being nearly overlapped in the Fig. 4c. The specific values can be found in Supplementary Table 6. In the "Steps" extraction task, GPT excelled with an accuracy of 0.966, recall of 0.972, and F1 score of 0.983. Finally, for the "Conditions" category, GPT achieved the highest accuracy (0.965), recall (0.972), and F1 score (0.981), indicating its robustness in this task. In comparison, the Phi model exhibited weaker performance on the majority of tasks. However, the GPT and Llama models showed relatively strong performance. Overall, GPT, Gemma, and Llama demonstrated impressive performance on various extraction tasks, with precision, recall, and F1 score metrics around 0.9, underscoring their reliability in information extraction. Baseline models such as MatBERT, SciBERT and MatSciBERT have achieved NER micro-F1 scores of 0.875, 0.734 and 0.861, and synthesis action retrieval scores of 0.717,0.701 and 0.919 respectively, as reported in the MatSciNLP benchmark[31,44]. In comparison, our approach achieves a NER F1 score of 0.962 and a synthesis action retrieval F1 score of 0.983, both of which significantly outperform these baselines.

LLMs, while exhibiting outstanding performance, suffer from hallucinations, especially on knowledge-intensive tasks. The hallucination of different LLMs was evaluated in the extraction of the complete synthesis route (comprising five tasks). The different definitions of hallucinations are shown in Fig. 5. As shown in Fig. 6, the results indicate that hallucinations do not frequently occur in the test articles. In analyzing the hallucinations of LLMs, the models may produce some hallucinations based on the text, except for those that are not explicitly described in the synthesis routes, such as "see supporting information" or "refer to synthesis methods of others". Even when these illusions occur, they are not entirely fabricated, but derived from textual information.

The sequence (logic) of experimental synthesis steps (actions) plays a crucial role, as it can greatly affect the properties and quality of the resulting material. Chemical reactions involved in different steps of the synthesis process are strongly dependent on the operation order. Incorrect sequencing can lead to incomplete reactions or the formation of undesirable by-products. The physical structure and properties of the material can be significantly influenced by the sequence of operations. The sequence of synthesis operations is crucial to the success of material synthesis, affecting everything from chemical reactions to reproducibility and safety. By understanding the sequence of synthesis actions, experimenters can more accurately replicate the process and achieve consistent material properties. The evaluation of the synthesis sequence provides invaluable insights for researchers.

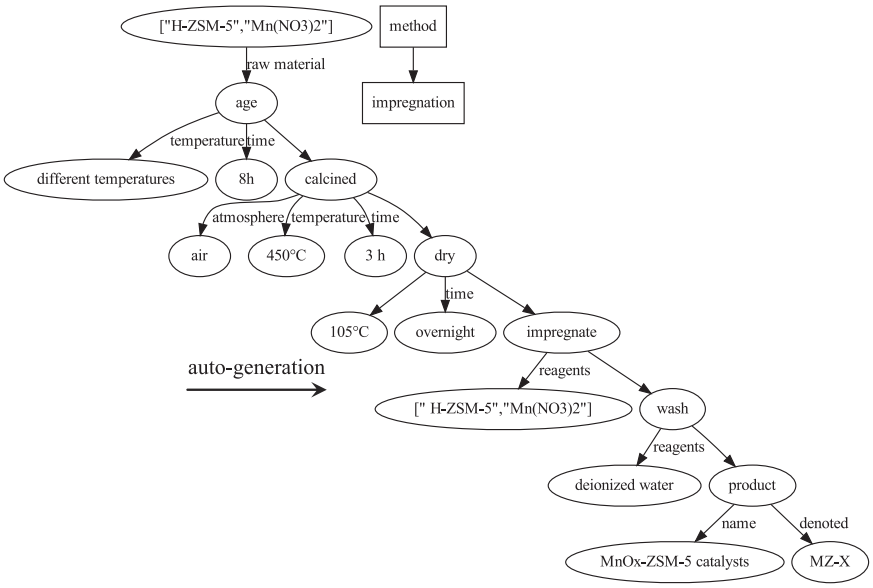


Fig. 9 | Catalyst synthesis and auto-generated structured flowchart. The flowchart contains preparation method, beginning with raw materials, followed by sequential synthesis steps with logically connected operations and associated reaction conditions (e.g., temperature, time, atmosphere). This process ultimately yields the final product.

The completeness and logicality of the synthesis steps of the four models in 20 articles were evaluated, and the results are shown in Fig.7. Among the four models, GPT has a significant advantage in the completeness of the number of synthesized actions extracted. The other three models exhibit incomplete extraction of steps in several articles. All four models perform well in the logic of the extracted steps, with all operation steps being presented in the correct order.

Extraction of material properties

The material synthesis process directly affects the microstructure of the material, and these microstructures determine the macroscopic properties of the material. Material properties can also be extracted using LLMs. One-shot prompt methods were employed to extract material properties such as activity, selectivity, and conversion of SCR catalysts. In the prompt, material properties, including conversion, selectivity, activity, and reaction conditions, were explicitly listed for extraction. Also, to minimize illusions, the model was instructed to output "not mentioned" for materials that did not have that property. Examples of one-shot prompts are in Supplementary Methods. Figure 8 shows an example of the inputs and outputs of LLM for extracting material properties and reaction conditions.

Large-scale data extraction

In our approach, the titles were extracted from the Extensible Markup Language (XML) nodes and then compared with a predefined list of titles related to material synthesis, such as "catalyst preparation" and "materials synthesis". To ensure robustness, the extracted titles were converted to lowercase and any leading or trailing whitespace was stripped, standardizing the titles for comparison. Next, the list of extracted titles was iterated and each title was compared with the predefined titles. If a match was found, the regular expression patterns were applied to the corresponding paragraph content to identify paragraphs discussing material synthesis. This process allowed for the effective extraction of relevant paragraphs from the literature. From this process, 2205 articles were identified containing paragraphs directly related to synthesis.

Following fine-tuning, the models are not required prompts in the input. They can directly process synthesis-related text to produce structured synthesis routes. Using the fine-tuned models, we performed large-scale extraction of synthesis information from these 2205 articles in the materials science literature. The models automatically generated structured synthesis routes and flowcharts from input synthesis paragraphs. Figure 9 shows an example of automatic flowchart generation from a paragraph[45]. These routes were processed and saved in a standardized JSON format, facilitating subsequent research applications and analyses. The standardized data can be accessed at https://github.com/Shaoruisun/LLMs-MatLitRecon.

As a result, a total of 48925 entities were successfully extracted, encompassing a wide range of information critical to understanding material synthesis processes. These entities include 3715 product entities, which represent the materials or compounds synthesized; 3186 preparation method entities, detailing the various methodologies employed; 3011 raw material entities, identifying the starting materials used in synthesis; 19102 synthesis action entities, which describe the specific steps involved in the synthesis processes; and 23626 preparation condition entities, outlining the conditions under which the synthesis was carried out, such as temperature, pressure, and time. The extraction of such a comprehensive set of entities enables a detailed and structured representation of the synthesis knowledge, facilitating further analysis and the potential for automation in material discovery and optimization.

Models transfer

Four LLMs, Llama, Gemma, Phi, and GPT, are fine-tuned to focus on materials science literature in the field of SCR. The fine-tuned models are intended to enhance their capability in extracting complex material synthesis routes and related information. Subsequently, these fine-tuned models were transferred to four other different domains, namely the HER, MSR, ORR, OER, and Lit-ion domains, in order to test their generalizability in different domains.

In each domain, 20 articles were randomly selected from the corresponding corpus. The DOIs of the test set articles are in Supplementary Table 7-11. Knowledge was extracted from the literature in each of the five domains using the fine-tuned models. Similarly, five tasks focusing on materials synthesis were performed: product extraction, raw material identification, synthesis method identification, extraction of synthesis steps, and generalization of synthesis conditions. The test results are presented in Fig. 10, with detailed results summarized in Supplementary Data (Supplementary Tables 12-15). As shown, the Llama, Gemma, and GPT models have excellent transferability, and the Phi model demonstrates slightly weaker performance. The results show that although different models perform distinctively on different domains and tasks, the fine-tuned models generally have high cross-domain transfer capabilities and are able to adapt to diverse materials science tasks.

Knowledge graph generation

Recent advances in materials informatics have demonstrated the critical role of ontologies and knowledge graph technology in enabling semantic,

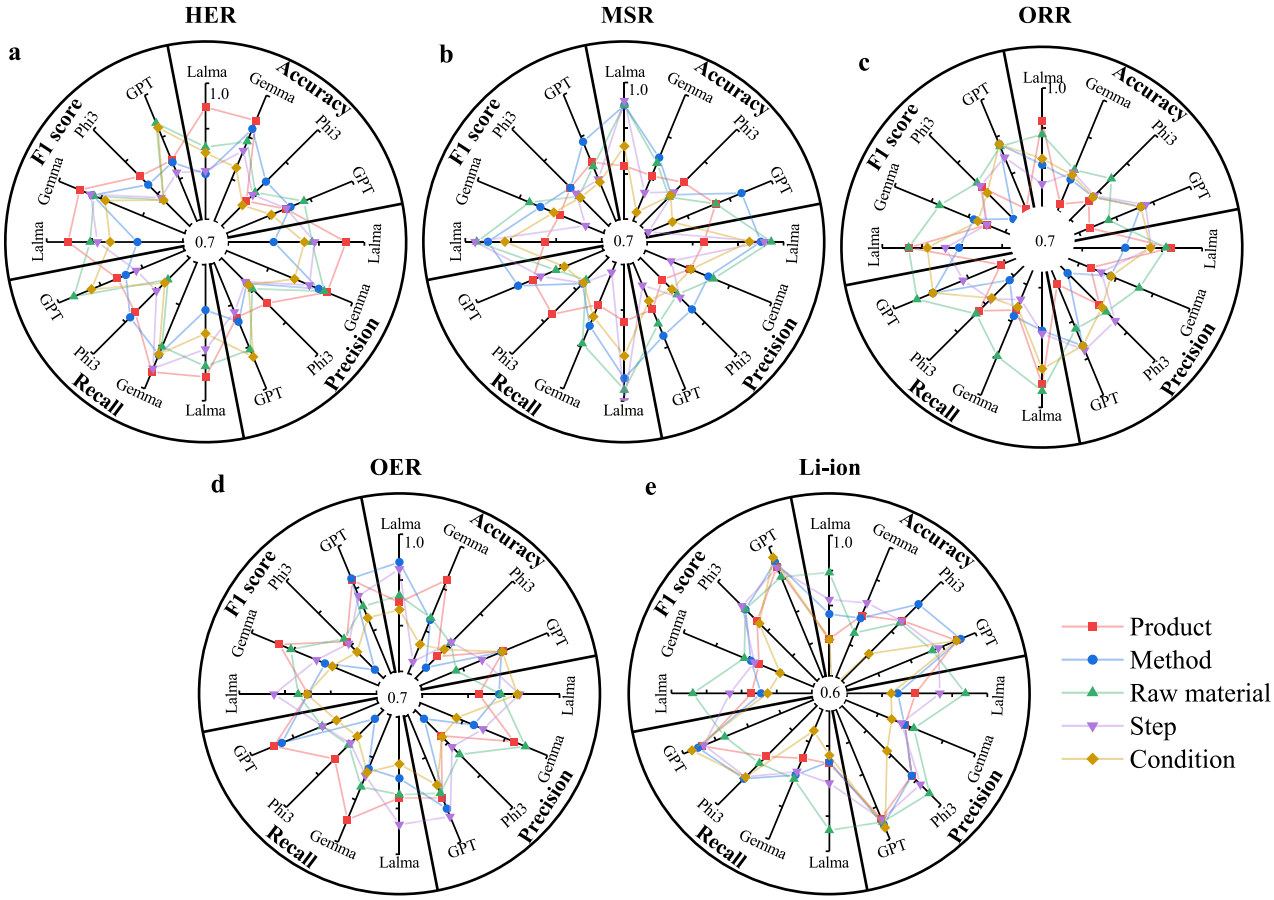


Fig. 10 | Performance of fine-tuned LLMs across five domains. The figure compares the performance of four LLMs (Llama, Gemma, Phi3, GPT) across five domains: **a** HER, **b** MSR, **c** ORR, **d** OER, and **e** Li-ion research. Model capabilities are evaluated using accuracy, precision, recall, and F1 score to assess their performance in extracting information related to products, methods, raw materials, steps, and conditions.

unambiguous, and interoperable data descriptions. Ontologies facilitate the transformation of data into machine-understandable Resource Description Framework (RDF) triples, enabling seamless integration of materials data and promoting interoperable exchange[46]. Knowledge is represented using the "subject-relation-object" structure (or triples) of "resource-property-value", serving as the structural foundation for machine semantic understanding. Moreover, interoperability between different knowledge graphs can be further enhanced by establishing mappings between semantic concepts[47]. In consideration of the FAIR principles, the extracted data are stored in JSON format, which is both human- and machine-readable. To ensure the structured and unified presentation of this information in the future, our work employs a structured, ontology-based template for synthetic information extraction, followed by constructing the knowledge graph to organize the information. The complete RDF dataset can be accessed at https://github.com/Shaoruisun/LLMs-MatLitRecon ("RDF dataset.xlsx")

The knowledge graph is a structured representation of information that captures relationships between entities, typically through a network of nodes (representing entities) and edges (representing relationships). Knowledge graphs are widely used to organize and integrate vast amounts of complex, interconnected data, facilitating both human understanding and machine processing. By leveraging triples in the form of (h,r,t), where "h" represents the head entity, "r" represents the relationship, and "t" represents the tail entity, knowledge graphs offer a formalized way to represent information through logical connections. The knowledge graph consists of a total of 26400 nodes and 55338 relationships. As illustrated in Fig. 11, the graph displays 1000 sets of relationships, offering a representative subset of the overall structure.

The knowledge graph was constructed with a hierarchical, catalyst-centered logic, as illustrated in the provided Fig. 12. On the left side, the graph presents a top-level structure where each catalyst, such as Ce/TiO_2, serves as the central node. Directly connected to this node are first-level entities, including raw materials used for synthesis, metadata such as literature sources (e.g., DOI, publication title, publication date, and other relevant data), key performance metrics (e.g., conversion), and general synthesis methods. This top-level view offers a concise summary of all primary information associated with the catalyst, allowing users to quickly grasp its background, preparation technique, and main application results. On the right side, the graph expands into a more detailed, multi-level structure by decomposing the synthesis route into a series of sequential experimental steps. The central catalyst node is linked to a synthesis step node, which aggregates all individual operations involved in the preparation

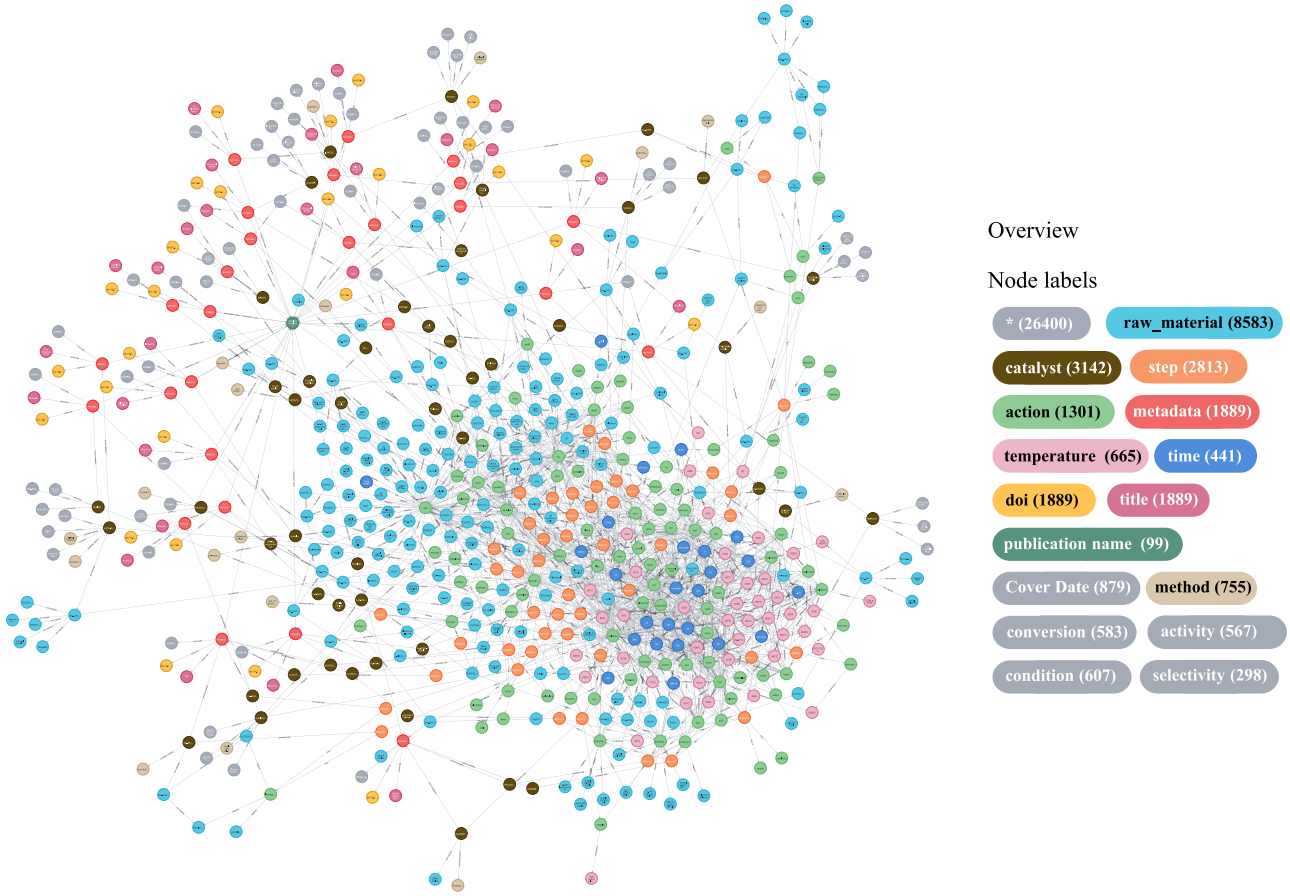


Fig. 11 | Visualization of partial synthetic entities and their relationships in the knowledge graph. The left panel depicts a subset of the knowledge graph, illustrating 1000 relationships among various synthesis entities, including raw materials, catalysts, process steps, temperature, and time. Nodes (colored circles) denote entities, while edges represent relational connections between them. The right panel provides the categorical classification legend, with numeric labels indicating the entity count per category.

process. Each of these steps—such as impregnation, dissolution, drying, calcination, grinding, and sieving—is represented as a separate node and is further annotated with associated reagents, processing conditions (such as temperature and time), and additional contextual or bibliographic information. This fine-grained organization allows users to precisely track the flow of materials and conditions throughout the entire synthesis, ensuring traceability and reproducibility.

This knowledge graph not only organizes and visualizes the complex relationships among catalysts, synthesis steps, and experimental conditions, but also provides an effective platform for retrieval. Specifically, it can visualize catalysts connected to various synthesis methods, enabling users to query catalysts according to their preparation techniques. Each cluster represents a group of catalysts associated with a specific synthesis method, with detailed nodes illustrating the relationships between these methods and the catalysts, as illustrated in Supplementary Discussion (Supplementary Fig. 1-8). Furthermore, this knowledge graph enables users to query specific preparation methods of interest, such as the wet impregnation method displayed in Fig. 13. Once a method is selected, users can further explore detailed information about the catalysts prepared. This capability allows for comprehensive analyses of how different synthesis strategies influence catalyst characteristics and overall outcomes, thereby facilitating data-driven research and hypothesis generation in the field of materials science.

The knowledge graph supports performance-based filtering of catalysts. As illustrated in Fig. 14, the graph allows users to rapidly identify high-performing materials, such as catalysts with a conversion rate exceeding 90%. For example, catalysts with high NO_X conversion efficiency were quickly highlighted within the graph. Researchers benefit greatly from this ability to filter catalysts based on specific performance metrics, including conversion rates and selectivity, which makes the search for promising candidates more efficient. A user guide for the knowledge graph is provided, demonstrating examples of filtering catalysts, preparation methods, raw materials, synthesis conditions, and performance metrics (https://github.com/Shaoruisun/LLMs-MatLitRecon, "Knowledge Graph Query Demo.mp4"). This guide allows for the efficient use of the knowledge graph in research applications, facilitating seamless data exploration and analysis. The ability to filter and identify key aspects, such as high-performance catalysts or specific synthesis techniques, is intended to accelerate research and development processes, promoting the discovery of new materials and the optimization of experimental designs. Furthermore, an interactive demonstration of the knowledge graph is available on GitHub (https://github.com/Shaoruisun/LLMs-MatLitRecon), allowing users to explore with these functionalities.

Conclusion

In this study, we developed a general framework for knowledge reconstruction from inorganic materials science literature. This framework demonstrates high accuracy and strong transferability across domains. The fine-tuned LLMs enabled precise and large-scale extraction of synthetic routes, performance, and metadata, creating a valuable structured dataset and a comprehensive knowledge graph. This work provides an effective method for organizing the vast amount of information in inorganic materials science literature, offering a structured data foundation for AI-driven acceleration of materials development. In future research, information on material mechanisms and characterization, as well as multi-modal data such

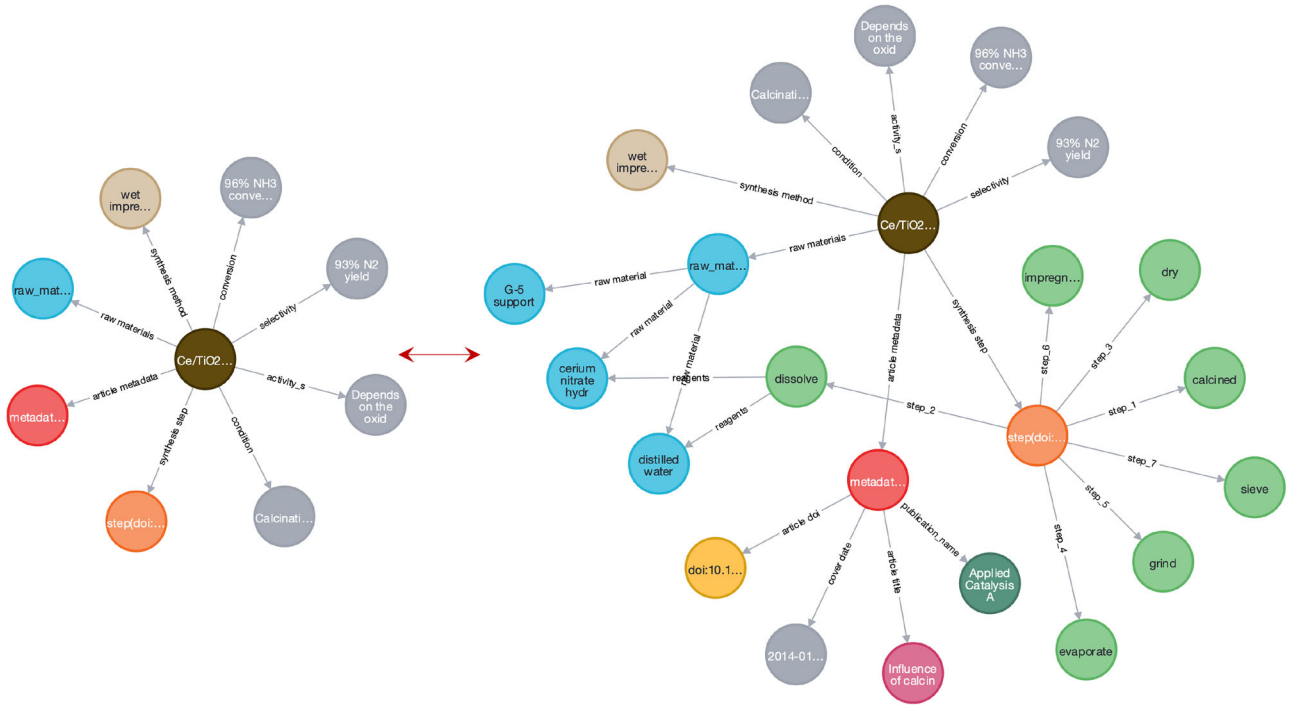


Fig. 12 | Example of a catalyst synthesis entities and relationships in knowledge graph. The left side shows the first-level nodes directly connected to the catalyst (Ce/TiO_2), including raw materials, synthesis steps, performance data (NH_3 conversion, N_2 yield), and metadata. The right side expands to show the second-level nodes, which detail further information such as preparation methods (e.g., wet impregnation), additional materials, and synthesis steps.

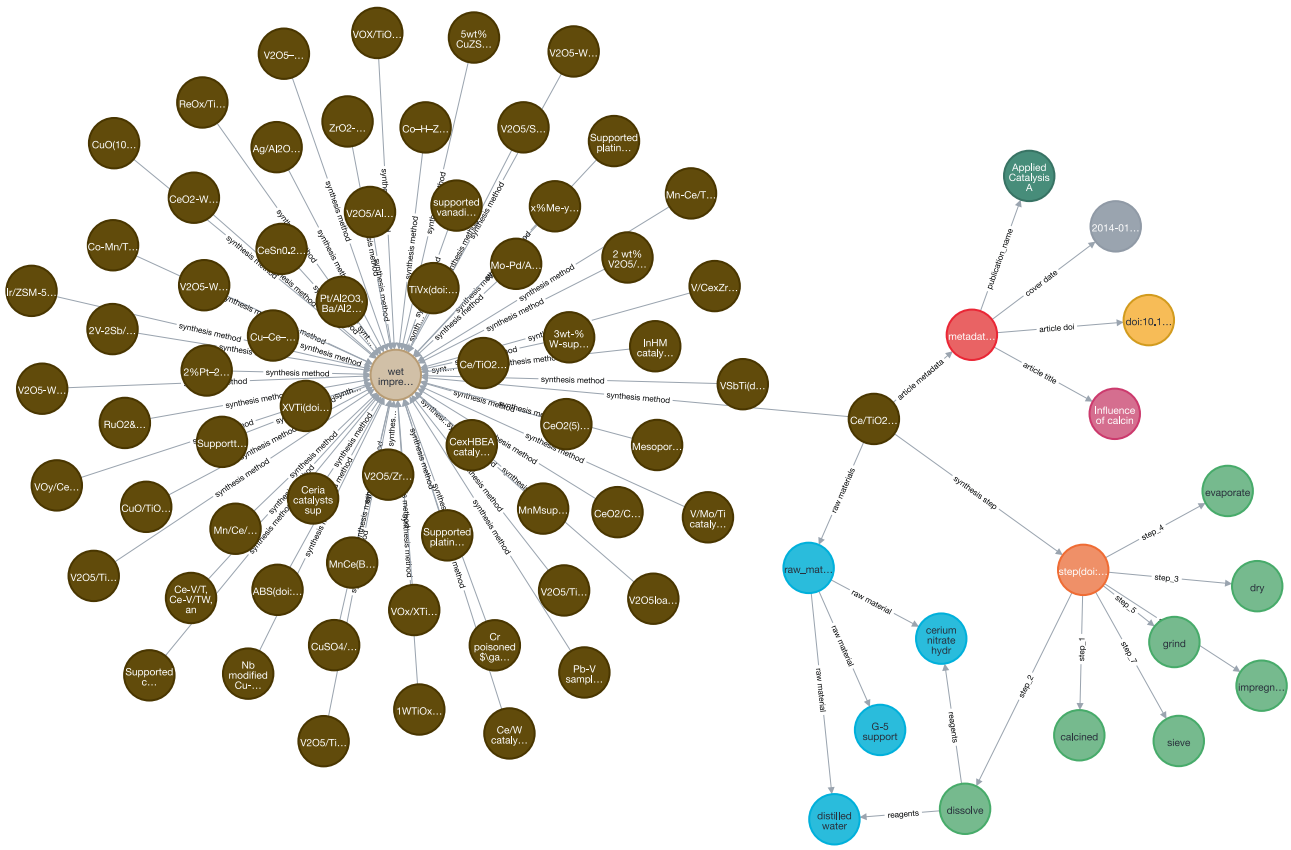


Fig. 13 | Knowledge graph of wet impregnation-based catalyst synthesis. The figure displays catalysts prepared using the wet impregnation method as the central node. It also shows connections to metadata, raw materials, and synthesis steps, allowing for detailed exploration of catalyst preparation and performance.

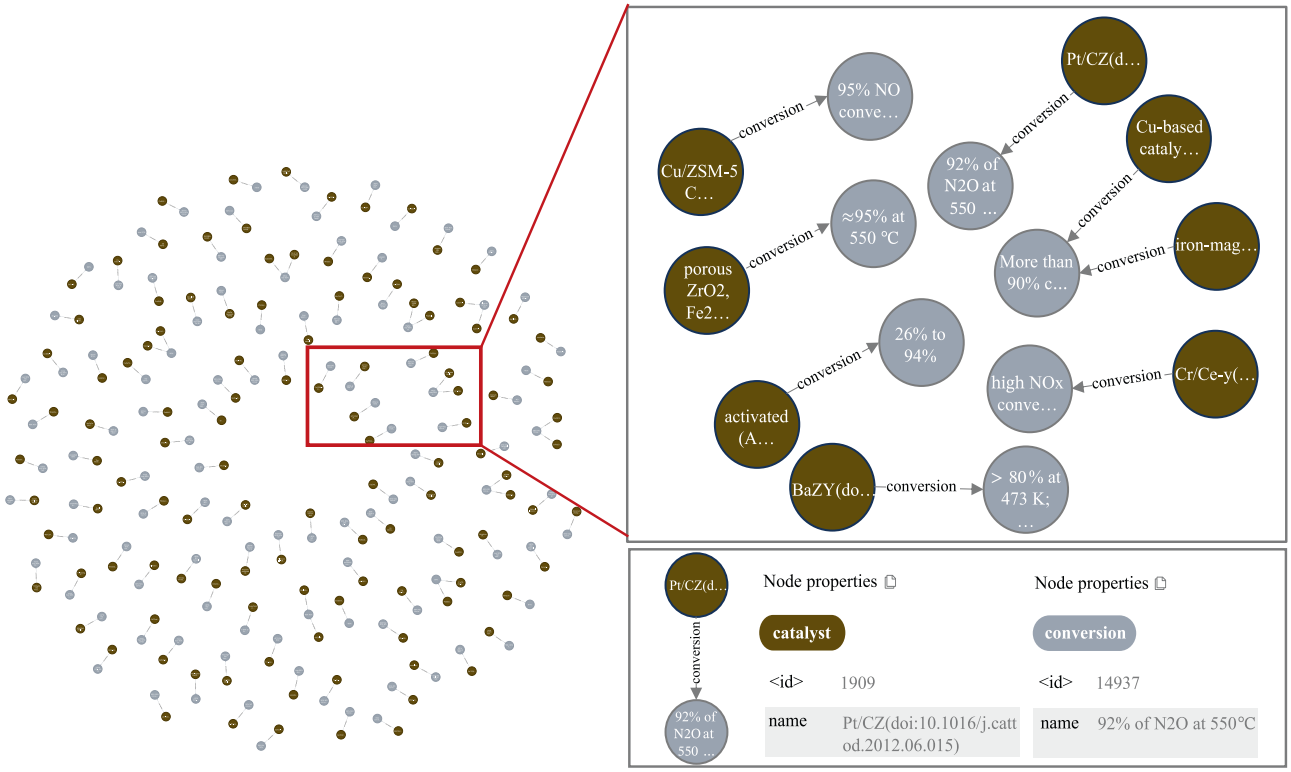


Fig. 14 | Knowledge graph filtering for high-performance catalysts. Filtered knowledge graph showing catalysts with conversion rates exceeding 90%. Brown nodes represent catalyst names, while gray nodes indicate associated conversion rates and reaction conditions. Magnified view of a local subgraph illustrating detailed node-specific information.

as images and spectra, can be systematically extracted and integrated. The knowledge graph can be further enriched with such information. This work could provide a foundation for data-driven materials discovery and design, promoting more efficient development of advanced materials.

Methods

Overall Procedure

A diagram outlining the general process for producing the fine-tuned model for material synthesis route is shown in Fig. 15. In the initial stage, a one-shot prompt, containing synthesis paragraphs and output styles, was designed as an example of prompt learning for the GPT-4 model. These extracted results were corrected and used as an initial training set for fine-tuning LLMs in the second stage. The third stage was an iterative training process. In this stage, new data were predicted, corrected, and added to the training set to update the fine-tuned model, thus improving its performance with each iteration. All input texts at each stage were selected from paragraphs that describe the material synthesis process, which were filtered from the original corpus. Finally, we employed fine-tuned LLMs to extract synthesis routes, and constructed a comprehensive flowcharts and knowledge graphs to illustrate the material synthesis process based on the synthesis routes. The extracted synthesis route from articles were presented as a JSON file containing. The extracted synthesis route data is publicly available at https://github.com/Shaoruisun/LLMs-MatLitRecon ("Fine-tuning data set.jsonl"). The application of LLMs in extracting and visualizing synthesis routes significantly enhances the accessibility and comprehensibility of material synthesis data, providing valuable insights for further research and development in this domain.

Datasets

The first stage aims to acquire a corpus of articles relevant to the topic. The relevant articles for constructing the training and testing datasets of this model were collected from the Elsevier database, under text and data-mining agreements. Several reasons make these journal articles particularly suitable for NLP workflows[48]. These articles can be downloaded and stored locally; they are digitized rather than scanned, at least when they are recent enough, so they can be processed without errors that might arise from having to convert images to text; they can be tracked via DOIs, so the data extracted from them can be referenced. Elsevier articles were downloaded via the ScienceDirect Search Application Programming Interface (API), which is provided by Elsevier for text and data mining purposes. We constructed text mining programs to identify and extract articles containing SCR syntheses. First, we repeatedly submitted the query text for "selective catalytic reduction" in the search bar of the Elsevier homepage via a Python web-scraping tool and recorded the responses on each page. Then, we extracted the metadata of the articles from the responses, including titles, author names, URL links, and DOI information. If a DOI or URL was provided, it was used to access the content of the relevant articles. Based on the extracted DOIs or URLs, the content of the article in XML format was further retrieved from the Elsevier server. Given that articles prior to 2000 were presented in PDF format, we limited the publication years to 2000−2023 when requesting articles from Elsevier, and downloaded a total of 12136 articles. The corpus contained 2305 articles with information on catalyst preparation paragraph. From these, 156 paragraph were selected for work-cleaned dataset construction based on two criteria: achieving a balanced distribution across publication years and ensuring a diverse representation of preparation methods.

To ensure high relevance to the research topic, articles containing specific keywords in their titles were retrieved using the same approach, which automatically downloaded literature from five additional categories related to the study. This process covered the period from 2000 to 2023, resulting in the creation of a comprehensive corpus of inorganic materials science literature. Specifically, 3009 papers on HER, 2401 on OER, 3460 on ORR, 428 on MSR, and 4052 on Li-ion batteries were downloaded. We developed an automated method using regular expressions to extract paragraphs discussing "preparation catalysts", "preparation materials", "synthesis catalysts", and "synthesis materials" from a large corpus of XML literature.

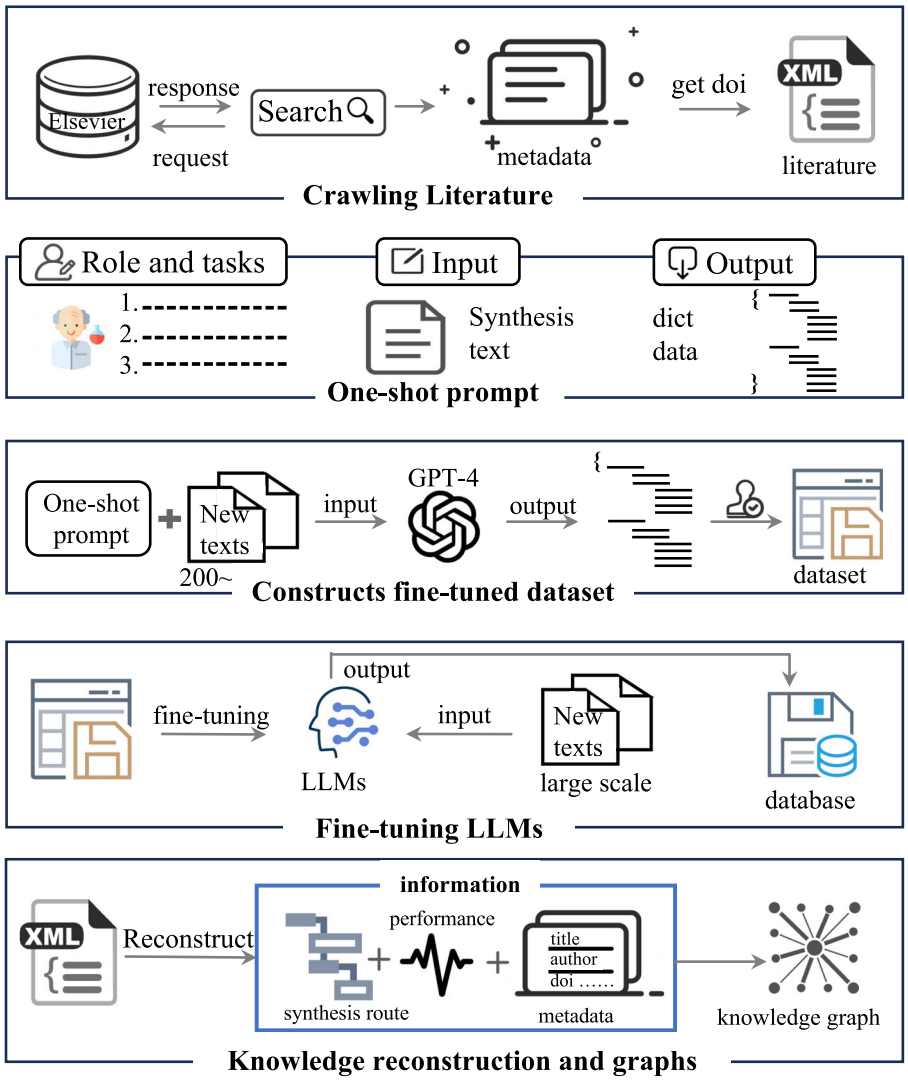


Fig. 15 | Workflow for information extraction using LLMs. This workflow covers literature crawling, one-shot prompting for data extraction, dataset construction, LLM fine-tuning, and knowledge graph generation from reconstructed literature.

The Output template of GPT-4

In recent years, the management and sharing of research data have become increasingly important in the materials science community. The FAIR principles have been widely recognized as fundamental guidelines for making research data more valuable and usable[49,50]. Applying these principles to materials synthesis information is critical to enable efficient data reuse, facilitate automated data integration, and accelerate data-driven material discovery. However, synthesis procedures are often reported in unstructured and inconsistent formats, limiting their accessibility and reusability. Following these principles, we introduced a structured template for inorganic materials synthesis routes, using a machine-readable JSON format. This approach ensures compatibility with the FAIR principles and promotes standardized data sharing within the community.

The structure and content of the synthesis are shown in Fig. 16. The synthesis routes are stored as JSON documents, which contain five components: products, synthesis methods, raw materials, synthesis steps, and corresponding synthesis conditions. The products include the name and denotation of the catalyst. All raw materials, including supports, reagents, solvents, etc., and their associated volumes (vol), concentrations (con), ratios, and/or masses were documented. All synthetic actions related to the synthesis steps were recorded in the precise order they were performed during the experimental synthesis. Additionally, the conditions associated with each synthetic action were noted, including temperature, time, pH, reagents, equipment, and atmospheric environment.

The documents follow a standardized format, with the exception of the synthesis steps, which differ depending on the specific requirements of each study. Unlike other sections, the synthesis steps are not predefined; instead, they are customized to accurately reflect the unique experimental procedures of each individual work. This approach ensures that the synthesis process is thoroughly documented, capturing all relevant details and variations that are critical for replicating and understanding the synthetic outcomes.

The prompt template for GPT-4

LLMs have the potential to extract additional or missed words, which presents several challenges for information extraction tasks. To address these challenges systematically, several potential measures can be considered. First, clarifying entity definitions and incorporating manual annotation review processes can help ensure extraction quality. Second, collecting standard terms, abbreviations, and entity variants from the field to build a structured vocabulary—such as a domain dictionary or knowledge graph—can enhance domain knowledge and support better contextual modeling. Third, automatic prompt generation algorithms could be adopted, which dynamically create guiding prompts based on the input text, reducing reliance on fixed templates. Fourth, for low-frequency terms,

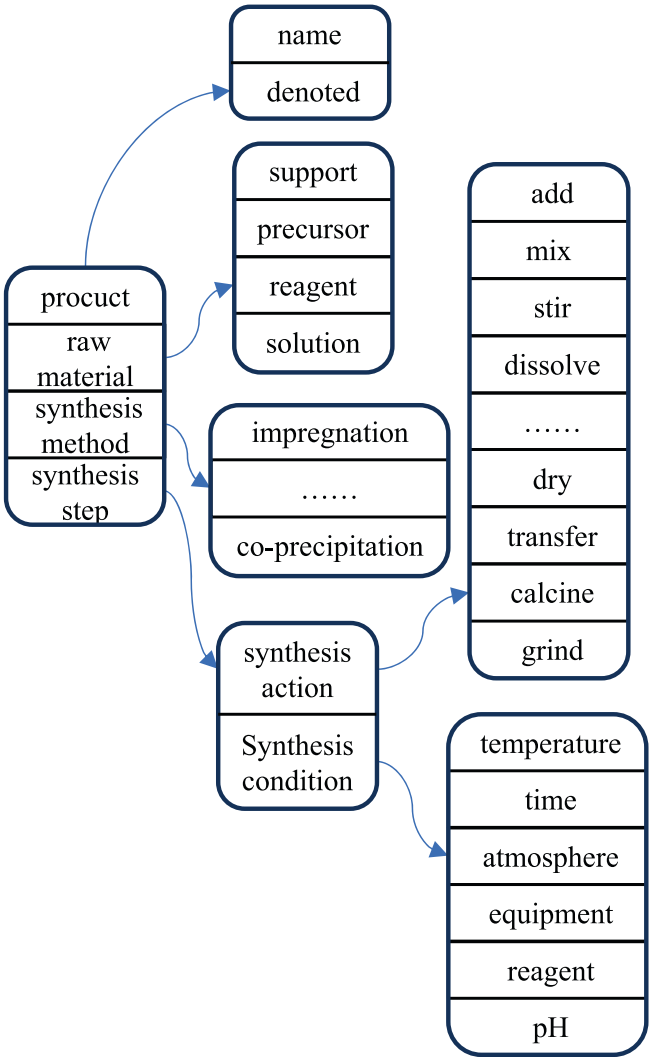


Fig. 16 | Schematic representation of the JSON template structure for inorganic material synthesis routes. It encompasses product details (name, denoted), raw materials (support, precursor, reagent, solution), synthesis methods (impregnation, co-precipitation, etc.), synthesis actions (add, mix, stir, dissolve, dry, transfer, calcine, grind, etc.), and synthesis conditions (temperature, time, atmosphere, equipment, reagent, pH).

few-shot learning approaches may be used where explicit prompt rules help guide entity extraction. Fifth, establishing domain-specific rules (such as regular expression matching or entity type constraints) can further filter and validate model outputs. These measures offer potential solutions for systematically overcoming the challenges related to extracting additional or missed entities with LLMs. In our work, we combined prompt-based method with human feedback to construct a domain-specific dataset, injecting domain knowledge into LLM and improving the information extraction capabilities.

The prompt template for GPT-4 consists of four main sections: role definition, task specification, input designation and output designation. It is designed as follows:

Role Definition: Role definition is used to define the role that the GPT-4 will play during the task and to set the context of the task.

For instance, GPT-4 is instructed to act as a scientific analyst, "Assume you are a chemical engineer specializing in the extraction of information from materials synthesis paragraphs.". This role determines the perspective and expertise employed by the model when processing tasks.

Task Specification: At this stage, the tasks will be described in detail. For the purpose of extracting materials synthesis route, this section clarifies the objective, such as identifying and summarizing the steps involved in the synthesis process, including the raw materials, methods, conditions and reagents used. This ensures that the model understands the scope and focus of the extraction task. We designed a total of four tasks:

1. Please output the synthesis information for the material based on the following paragraph, including the product name, preparation method, raw materials, and all steps.

2. The steps are operational actions, such as addition, stirring, mixing, dissolving, transferring, heating, drying, calcining, reducing, activating, and other similar operations.

3. Output each operation in order, and the conditions associated with each operation, such as temperature, time, reagents, equipment and instruments, atmosphere, etc.

4. Please extract the synthesis information separately and output it in dictionary format.

Input template: The input consists of paragraphs detailing the synthesis of materials. To illustrate, an example synthesis paragraph is provided within the prompt template. This paragraph was selected because it contains a comprehensive synthesis procedure with detailed descriptions of each step.

Output template: In the prompt, the output is the synthesis routes extracted from the input paragraphs, following the predefined template. At first, the example synthesis routes will be formatted according to the preset template and presented the required information. Subsequently, the pre-trained model will autonomously generate synthesis routes based on new input passages according to the same structural format. The one-shot prompt is shown in Fig. 17, and the full information is provided in Supplementary Methods.

LLMs

In this study, we fine-tuned several models downloaded from Hugging face using the LLaMA-Factory framework. To conserve computational resources, we opted for smaller parameter versions of each LLM. The models included Gemma (Google), Llama (Meta), and Phi (Microsoft). Additionally, we utilized an online GPT model provided by OpenAI. A detailed description and hyperparameters of each model are given below.

The comprehensive overview is provided of three open-source LLMs: Gemma, Llama, and Phi. Each model is designed to cater to various computational needs that enhance their performance on consumer-grade hardware.

Gemma is a series of lightweight, state-of-the-art open models from Google, with two versions (2B and 7B). Each version includes a pre-trained foundational model and an instruction-optimized model, and is capable of running on consumer-grade hardware without data quantization for up to 8 K tokens. We utilized the 7B parameters pre-trained foundational version.

Meta's Llama 3, the latest in the Llama series, offers 8B and 70B versions. Each version has a base model and an instruction-tuned model. We use the 8B parameter instruction-tuned version (LLaMA3-8B-instruct) with a context length of 8 K tokens.

Phi-3 is a series of open AI models by Microsoft. We fine-tuned the Phi-3-mini model, a language model with 3.8 billion parameters and a context length of 128k tokens.

All three models were fine-tuned using the LLaMA-Factory framework with the LoRA method. The fine-tuning process parameters involved setting the training batch size to 1, gradient accumulation steps to 8, learning rate to 3.0e-5, number of training epochs to 24, logging steps to 1, and save steps to 50.

For the GPT model, we utilized the GPT-3.5-turbo-1106 from OpenAI. The hyperparameters for fine-tuning this model are different from those used for the Hugging face models. The batch size was set to 1, with a learning rate multiplier of 1, within the recommended range of 0.1 to 10, and the model was fine-tuned over 3 epochs. These parameters were chosen to ensure optimal performance and efficient training for the GPT-3.5-turbo model.

Evaluate method

The performance of the LLMs were evaluated on the basis of precision, recall and F1 score. Precision is the fraction of correct (relevant) records among all extracted records; recall is the fraction of successfully extracted records

GPT-4 Prompt Demo

Request: " Assume you are an engineer specializing in the extraction of information from materials synthesis paragraphs. 1:Please output the synthesis information for the material based on the following paragraph, including the product name, preparation method, raw materials, and all steps. 2:The steps are operational actions, such as addition, stirring, mixing, dissolving, transferring, heating, drying, calcining, reducing, activating, and other similar operations. 3:Sequentially output the operation corresponding to each step and conditions related to it such as temperature, time, reagents, equipment and instruments, atmosphere, etc. 4:Please extract the synthesis information separately and output it in a dictionary format."

Context: "Catalyst synthesis Fe/TiO2was prepared using an F127-assisted sol-gel process. In a typical preparation, F127 (0.4 mmol) was added to sufficient anhydrous ethanol and the solution was heated gently until the F127 was completely dissolved. Then certain amounts of TBOT and CH3COOH were dissolved in the F127 solution under vigorous stirring for 30 min, giving a transparent yellow solution. Fe(NO3)3·9H2O solution was then added drop-wise to the solution under continuous stirring…….
For comparison, supported Fe/TiO2catalysts were also prepared using wet impregnation and coprecipitation methods. In the impregnation method, anatase TiO2(Hehai Technology Company, China) was impregnated with an appropriate amount of Fe(NO3)3·9H2O solution. After stirring for 30 min, the mixture was dried overnight at 110 °C and then calcined at 450 °C in air for 3 h. The Fe/TiO2catalysts prepared by impregnation and coprecipitation were denoted by Fe/TiO2(i) and Fe/TiO2(c), respectively."

```
answer_dict = {
  "Catalyst1": {
    "product": {
      "name": "Fe/TiO2",
      "denoted": "Fe/TiO2(t)"
    },
    "method": "F127-assisted sol-gel",
    "raw material": "F127",
    "step": {
      "add1": {
        "reagents": ["F127", "anhydrous ethanol"]
      },
      "heat": [],
      "dissolve": {
        "reagents": ["TBOT", "CH3COOH"]
      },
      "stir1": {
        "time": "30min"
      },
      "add2": {
        "reagents": "Fe(NO3)3·9H2O"
      },
      "stir2": {
        "time": "2h"
      },
      "keep": {
        "temperature": "room temperature",
        "time": "3d"
      },
      "dry": {
        "temperature": "110 °C",
        "time": "12 h",
        "instrument": "oven"
      },
      "calcined": {
        "temperature": "450 °C",
        "time": "3 h"
      }
    }
  },
  "Catalyst2": {……},
  "Catalyst3": {……}
}
```

Fig. 17 | One-shot prompt learning designed for GPT-4 to construct a work-cleaned dataset. This figure shows a GPT-4 prompt demo. It includes a request for extracting material synthesis information (covering product name, method, raw materials, steps with operations and conditions), and the dictionary - format output of extracted synthesis details for catalysts.

among all correct (relevant) records in the articles; and F1 score is the harmonic mean of precision and recall, defined as

$$accuracy = \frac{TP + TN}{FP + FN + TP + TN}$$

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

$$F1score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

where TP is the true positive count (the number of records that are correctly extracted), FP is the false positive count (the number of records that are

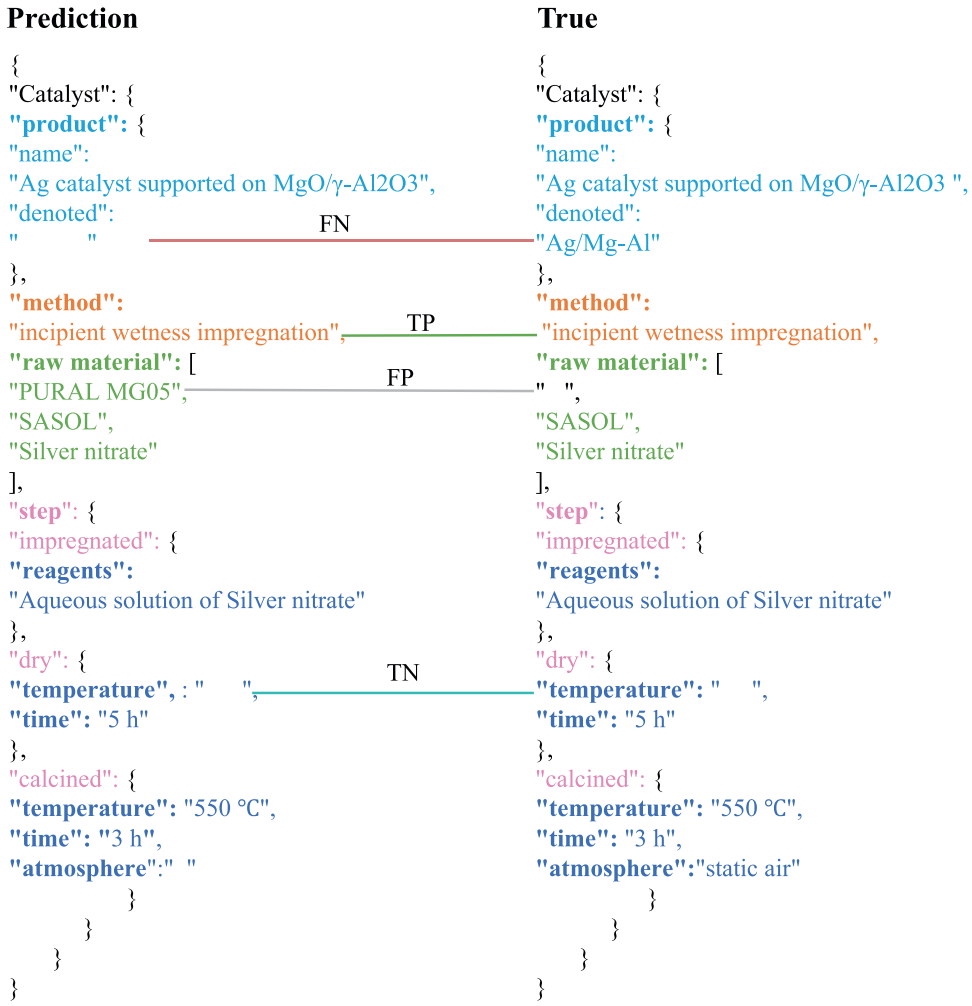


Fig. 18 | Evaluation methods for model outputs. Schematic illustration of defining evaluation metrics (True Positive (TP), False Negative (FN), False Positive (FP), True Negative (TN)) by comparing predicted and true catalyst synthesis information.

incorrectly extracted), and FN is the false negative count (the number of correct records that are not extracted). Their definitions are shown in Fig.18.

Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files. The structured datasets supporting the findings—including the corpus information, fine-tuning data, the structured data, knowledge graph triples and the code—are publicly available in the GitHub repository at https://github.com/Shaoruisun/LLMs-MatLitRecon. Any other relevant data are available from the corresponding author Shaorui Sun, email: sunsr@bjut.edu.cn, on reasonable request.

Code availability

The custom code used to process, and analyze the data in this study is publicly available in the GitHub repository at https://github.com/Shaoruisun/LLMs-MatLitRecon. No restrictions apply to access the code.

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Author contributions

S.L. contributed the research design, developed the computational framework, conducted experiments, analyzed results, and wrote the initial manuscript. S.W. assisted in fine-tuning the models and processed data. C.H. contributed to statistical analysis and data validation. Y.Z. participated in data validation and reviewed the manuscript. G.Z. contributed to manuscript revision, co-supervising the research process. S.S. supervised the project, provided critical resources, and acquired funding. S.S. also contributed to manuscript revision and overall research coordination.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Additional information

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